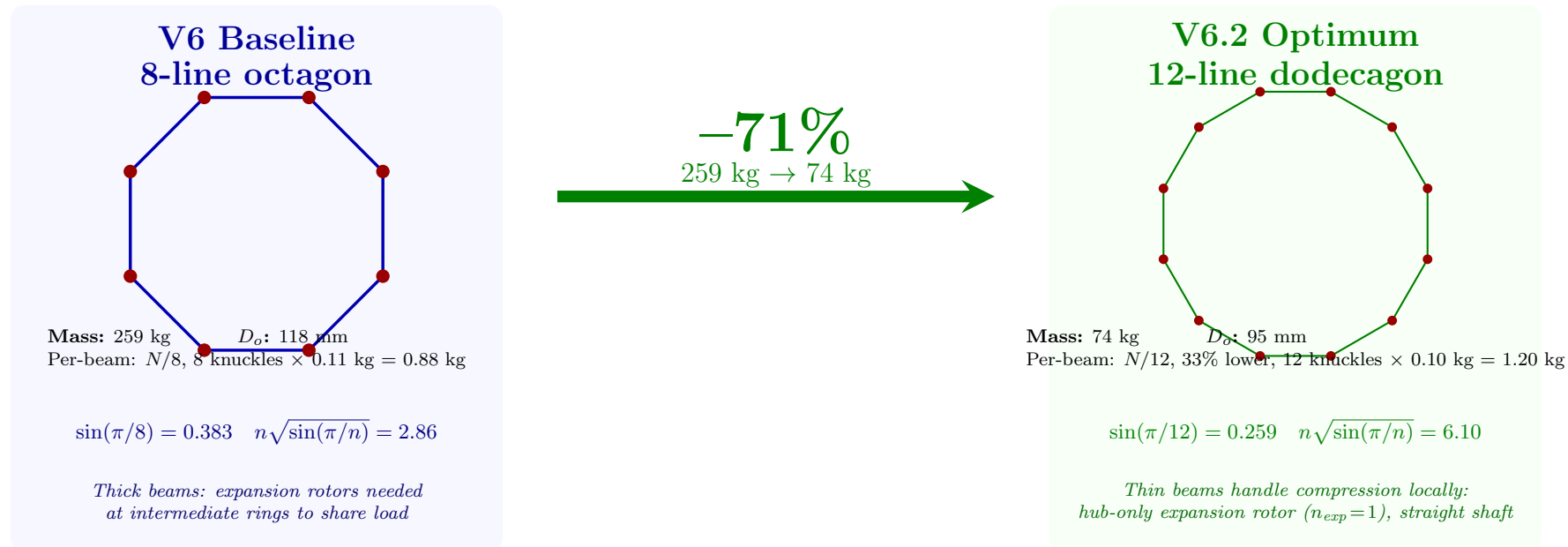


The Polygon Flip: Why 12 Lines Beat 8

Beam-only scaling says fewer lines are lighter. The DE optimiser found the opposite.



What this reveals about the TRPT scaling problem:

Beam-only model: $m_{\text{ring}} \propto n\sqrt{\sin(\pi/n)}$ is monotonically increasing — $n = 12$ costs 2.19 $\times$ more beam mass than $n = 3$. That suggests fewer lines are lighter. **But the beam-only model ignores the coupled system.** Thinner beams (95 vs 118 mm OD) enable smaller knuckle hardware, and knuckle mass scales with beam diameter. Tether mass ($\propto n$) and rotor solidity penalties are modest. **The DE optimiser evaluates the full coupled design space** — beams, knuckles, tethers, rotor sizing, ring count, density profile, expansion stations — and finds the global minimum at $n=12$, not $n=3$. This is a system-level inversion of a component-level intuition.